

Architectural Considerations for AI-Assisted Development at NOAA EMC

A Comparative Analysis of LLM Integration Strategies

Enterprise Infrastructure Branch (EIB)

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Bottom Line Up Front (For Leadership)

What EIB needs to acquire:

1. **IDE-integrated AI coding assistant licenses** — provides in-editor code completion, chat, and code generation for developer productivity
2. **FedRAMP-authorized LLM API access** (pay-per-use) — enables programmatic model calls for our custom MCP-RAG system integration and automation pipelines

These are complementary: IDE assistant for daily developer productivity, API access for infrastructure integration. Both integrate with our existing NOAA-specific knowledge system (MCP-RAG).

Executive Summary

NOAA EMC requires AI-assisted coding capabilities that integrate with developer workflows while meeting federal security requirements. This document clarifies a critical architectural distinction: **the Agentic Software Development system is not the same as the Large Language Model (LLM) that powers it.**

Our MCP-RAG platform (currently operational) provides the *domain-specific intelligence layer*—knowledge of Global Workflow structure, EE2 compliance rules, and HPC configurations. This layer is **LLM-agnostic by design**. Combined with GitHub Copilot’s emerging multi-model support (GPT-4, Claude, and others), NOAA gains:

1. Access to multiple LLMs through a single GitHub Copilot license
2. Domain-specific context via MCP tools that no generic LLM possesses
3. Flexibility to leverage new models as they become available

The question before leadership is: which procurement pathway provides the best combination of FedRAMP compliance, model flexibility, and integration with our existing MCP-RAG infrastructure?

The Architectural Demarcation

Key Insight: The foundation model (LLM) is a commodity input.
The differentiated value is in the *context layer* that makes it useful for NOAA.

Layer 1: The LLM (Interchangeable)

Foundation models from OpenAI, Anthropic, Google, and AWS all provide similar core capabilities: code generation, natural language understanding, and reasoning. They compete on price, speed, and benchmark performance—metrics that shift quarterly.

Layer 2: The MCP-RAG System (NOAA-Specific)

Our system provides what generic LLMs cannot:

- **Structural Knowledge:** Neo4j graph database mapping 86,000+ code relationships in Global Workflow
- **Semantic Knowledge:** ChromaDB vector store with 15,000+ documents (EE2 standards, HPC procedures, Fortran best practices)
- **Compliance Enforcement:** Automated EE2/NCO standards validation
- **Platform Awareness:** Configurations for Hera, Hercules, Orion, WCOS2

Layer 3: The IDE Integration (Flexible)

The developer interface (VS Code, Cursor, Windsurf) connects to Layer 2 via the Model Context Protocol (MCP). The IDE choice is independent of both the LLM and the knowledge layer.

LLM Access: GitHub Copilot vs AWS Bedrock

A critical distinction exists between *IDE coding assistance* and *programmatic API access*. AWS Bedrock provides both a chat interface and full API access, while GitHub Copilot provides IDE assistance only. These serve fundamentally different purposes:

GitHub Copilot (IDE Assistance Only):

- Provides coding assistance within VS Code (autocomplete, chat, code generation)
- Subscription (\$10–39/month) includes multi-model access (Claude, GPT-4, Gemini)¹
- **Does NOT provide API keys** for custom application development
- **Cannot be used for:** RAG ingestion, embedding generation, custom inference pipelines, or programmatic integration with MCP-RAG systems

¹GitHub Copilot multi-model access:

<https://docs.github.com/en/copilot/using-github-copilot/using-claude-sonnet-in-github-copilot>

AWS Bedrock (Full API Access):²

- Provides programmatic API access to foundation models (Llama, Titan, Mistral, Claude 4.x*)
- **Enables custom development:** RAG systems, embedding generation, inference pipelines
- Can integrate with ChromaDB, Neo4j, and MCP-RAG infrastructure
- FedRAMP High authorized (GovCloud)
- *Claude 4.x requires separate Anthropic EULA beyond the base Bedrock agreement

Implication for EIB Development: GitHub Copilot is valuable for day-to-day coding assistance, but our MCP-RAG system development requires actual API access that only AWS Bedrock (or direct provider APIs) can provide. These are complementary, not competing, capabilities. The table below reflects this nuance: models requiring separate vendor agreements are noted accordingly.

Option	Pros	Cons
OpenAI API (GPT-4/4o)	• FedRAMP Moderate authorized • Mature API with extensive documentation • GitHub Copilot can use this backend • Strong code generation benchmarks	• Single-vendor lock-in risk • Pricing volatility (3 changes in 2024) • Limited transparency on training data • API rate limits can throttle teams
Anthropic Claude (Claude 4.x/Opus)	• Superior code reasoning (SWE-bench leader) • Native MCP support (Anthropic created MCP) • Available via AWS Bedrock (with EULA) • Strong safety/alignment track record	• No direct FedRAMP authorization • Direct procurement requires commercial agreement • Bedrock pathway still requires Anthropic EULA • Smaller enterprise footprint than OpenAI

²AWS Bedrock API access documentation:

<https://docs.aws.amazon.com/bedrock/latest/userguide/getting-started-api-keys.html>

Option	Pros	Cons
Google Gemini (Gemini Pro/Ultra)	• FedRAMP Moderate (Google Cloud) • Competitive pricing • Strong multimodal capabilities • Existing NOAA Google relationships	• Exclusive Gemini procurement would have less mature IDE integration than GitHub Copilot or Amazon Q Developer VS Code plugins • MCP requires custom adapter • Coding benchmarks lag Claude/GPT-4 • Enterprise support tier required
AWS Bedrock (Multi-model)	• FedRAMP High (Gov-Cloud) • Access to Llama, Titan, Mistral (included) • Claude 4.x/Opus available (with EULA) • Amazon Q Developer VS Code plugin integrates with EIB MCP/RAG • Pay-per-token, no seat licenses	• Claude access requires separate Anthropic EULA • Higher latency than direct API • Requires AWS account infrastructure • Smaller IDE plugin ecosystem than GitHub Copilot
Azure OpenAI (GPT-4 via Azure)	• FedRAMP High (Azure Gov) • Same GPT-4 models as OpenAI direct • Enterprise SLAs and support • Data residency controls	• Requires Azure subscription commitment • Additional abstraction layer • Model availability lags OpenAI by weeks • Higher cost than direct OpenAI API

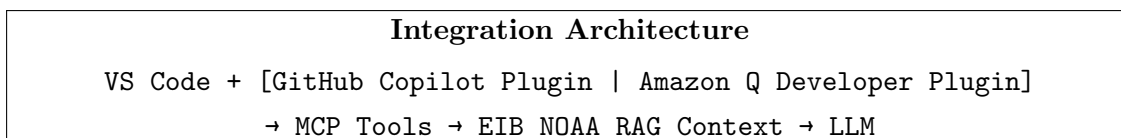
Addressing Key Concerns

“We need coding assistants that integrate into our IDEs”

The IDE integration landscape has matured significantly. Both **GitHub Copilot** and **Amazon Q Developer** function analogously as AI coding assistants—both have VS Code plugins that integrate with our EIB MCP-RAG system.

- **GitHub Copilot:** VS Code plugin supporting multiple models (GPT-4, Claude, others). Model selection available in settings. Integrates with MCP. Also offers **Copilot CLI** for terminal assistance, which integrates with the VS Code integrated terminal.
- **Amazon Q Developer:** AWS’s VS Code plugin using Bedrock models. FedRAMP High via GovCloud. Integrates with MCP.
- **Claude CLI:** Anthropic’s terminal-based coding assistant. *Note: GitHub Copilot CLI also integrates seamlessly with the VS Code integrated terminal, offering equivalent capabilities. Additionally, Claude models can be selected within GitHub Copilot’s multi-model offering.*
- **Google Gemini Code Assist:** Uses Gemini models only.

Both **GitHub Copilot** and **Amazon Q Developer** integrate with our **EIB MCP-RAG system** via their respective **VS Code plugins**, providing developers access to NOAA-specific domain knowledge regardless of which platform is chosen.



The MCP layer provides the *domain-specific context* (NOAA workflows, EE2 standards, HPC configurations) that makes responses accurate for NOAA-specific queries—independent of which IDE plugin is used.

“We’re interested in deployment, not development of GenAI”

Correct distinction. AWS Bedrock and similar services are *deployment platforms* for inference, not model training. They are appropriate for our use case: sending prompts, receiving completions, and integrating responses into developer workflows.

AWS Bedrock and Amazon Q Developer

AWS Bedrock is an API service for calling foundation models, while **Amazon Q Developer** is AWS’s IDE plugin—analogous to GitHub Copilot. Both integrate with NOAA’s MCP-RAG system.

Component Relationship

Amazon Q Developer Capabilities (January 2026)

- Inline code suggestions with vulnerability scanning
- Agentic capabilities: feature implementation, testing, code review, refactoring
- Java upgrade automation (8→17) and .NET porting (Windows→Linux)
- CLI integration with terminal autocompletions
- Available in AWS Console, Microsoft Teams, Slack, and GitHub Enterprise
- **Free tier:** 50 agentic chat interactions/month; Pro tier for enterprise features
- **FedRAMP High** authorization via AWS GovCloud

Component	Role
AWS Bedrock	API service providing access to foundation models (Llama, Titan, Mistral, Claude 4.x*).
Amazon Q Developer	AWS’s VS Code plugin (also available for JetBrains, Visual Studio, Eclipse). Provides code completions, chat, and agentic capabilities. Works analogously to the GitHub Copilot VS Code plugin.
MCP Integration	Both the GitHub Copilot and Amazon Q Developer VS Code plugins integrate with our EIB MCP-RAG system, providing domain-specific NOAA context to either platform.

Setup Considerations

Amazon Q Developer requires AWS IAM credential configuration. GitHub Copilot authentication is managed through NOAA-enabled GitHub accounts, facilitated by Microsoft’s enterprise SSO infrastructure. Both approaches are standard enterprise authentication patterns. For organizations with existing AWS infrastructure, Q Developer integrates naturally with existing IAM policies and GovCloud environments.

“The Anthropic \$1 deal is a no-go”

Direct Anthropic procurement is blocked by FedRAMP requirements. AWS Bedrock now offers the full Claude 4.x model family (including Opus 4.5), but accessing them still requires a separate End User License Agreement (EULA) with Anthropic—meaning the procurement pathway remains complex. GitHub Copilot also offers Claude model access, which may provide a simpler procurement path depending on enterprise licensing arrangements.

Strategic Recommendation Framework

Rather than selecting a single provider, NOAA should evaluate options against these weighted criteria:

Criterion	Weight	Notes
FedRAMP Authorization Level	30%	High preferred; Moderate acceptable
MCP/RAG Integration Compatibility	25%	Must work with existing infrastructure
Code Generation Quality	20%	SWE-bench, HumanEval benchmarks
Total Cost of Ownership	15%	Licensing + infrastructure + support
Vendor Stability/Lock-in Risk	10%	Multi-model options reduce risk

Table 2: Suggested evaluation weighting for LLM provider selection

Architecturally Sound Options

Based on current FedRAMP status and integration requirements, the viable paths are:

1. **GitHub Copilot Enterprise:** Multi-model access (GPT-4, Claude) through familiar IDE integration. FedRAMP status depends on GitHub Enterprise Cloud authorization. Integrates with MCP-RAG system.
2. **AWS Bedrock + Amazon Q Developer:** FedRAMP High via GovCloud. Access to Llama, Titan, Mistral without additional agreements. The Amazon Q Developer VS Code plugin works analogously to the GitHub Copilot VS Code plugin and integrates with our EIB MCP-RAG system.
3. **Azure OpenAI:** If organization has existing Azure commitment. FedRAMP High with GPT-4 access. Enterprise SLAs and support.
4. **Google Gemini (Google Cloud):** Viable if NOAA expands Google relationship. FedRAMP Moderate. Requires MCP adapter development.

Conclusion

The foundation model market continues to evolve rapidly. NOAA’s strategic advantage lies in the **domain-specific MCP-RAG infrastructure** that provides institutional knowledge no generic LLM possesses—and this infrastructure integrates with both GitHub Copilot and Amazon Q Developer.

The procurement decision should prioritize:

1. **FedRAMP compliance** at the required authorization level (High vs. Moderate)
2. **Existing infrastructure alignment** (AWS vs. GitHub/Microsoft ecosystem)
3. **Model flexibility requirements** (single model vs. multi-model access)

Both GitHub Copilot and AWS Bedrock (via the Amazon Q Developer VS Code plugin) provide viable paths that integrate with our EIB MCP-RAG system. The choice depends on organizational infrastructure preferences and FedRAMP authorization requirements.

*Prepared by: Enterprise Infrastructure Branch (EIB), NOAA Environmental Modeling Center
For questions: Contact EIB AI Infrastructure Team*